Overview of Health Informatics

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INTRODUCTION: Why Medical/Health Informatics Evolved?

- Due to the multiple challenges facing the practice of medicine today.
- As an example, clinicians need to:
  - be more efficient,
  - migrate away from paper based-records,
  - reduce medication errors and have educational and patient related information at their fingertips.

Chapter 1, Hoyt, R.E., Sutton, M. and Yoshihashi, A., 2008. *Medical Informatics: Practical Guide for the Healthcare Professional 2008*. Lulu. com.

Definitions

- application of computers, communications and information technology and systems to all fields of medicine - medical care, medical education and medical research"
- "understanding, skills and tools that enable the sharing and use of information to deliver healthcare and promote health"

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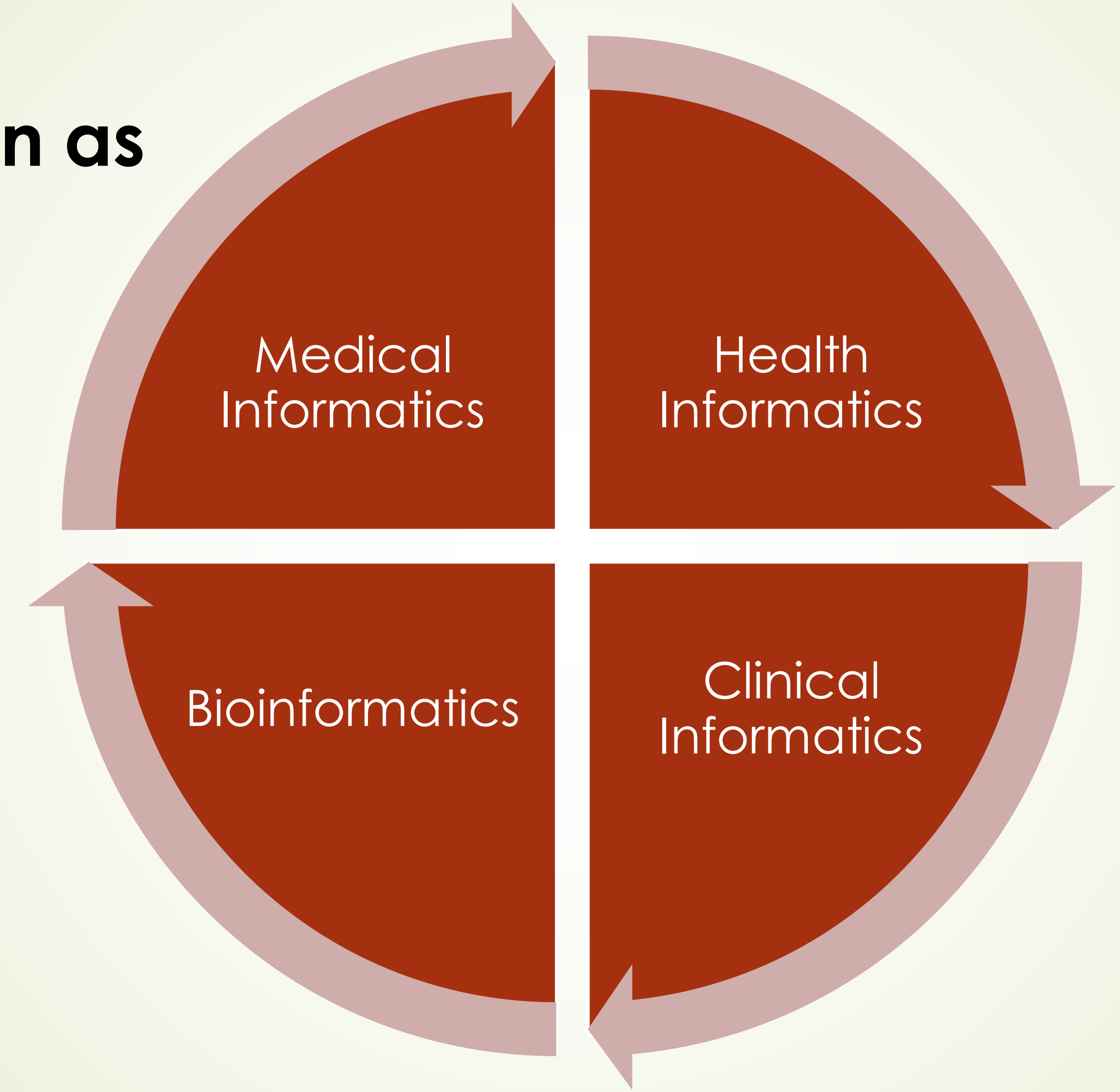
also known as



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- **Bioinformatics** involves the integration of biology and technology and can be defined as the:
- “analysis of biological information using computers and statistical techniques; the science of developing and utilizing computer databases and algorithms to accelerate and enhance biological research”

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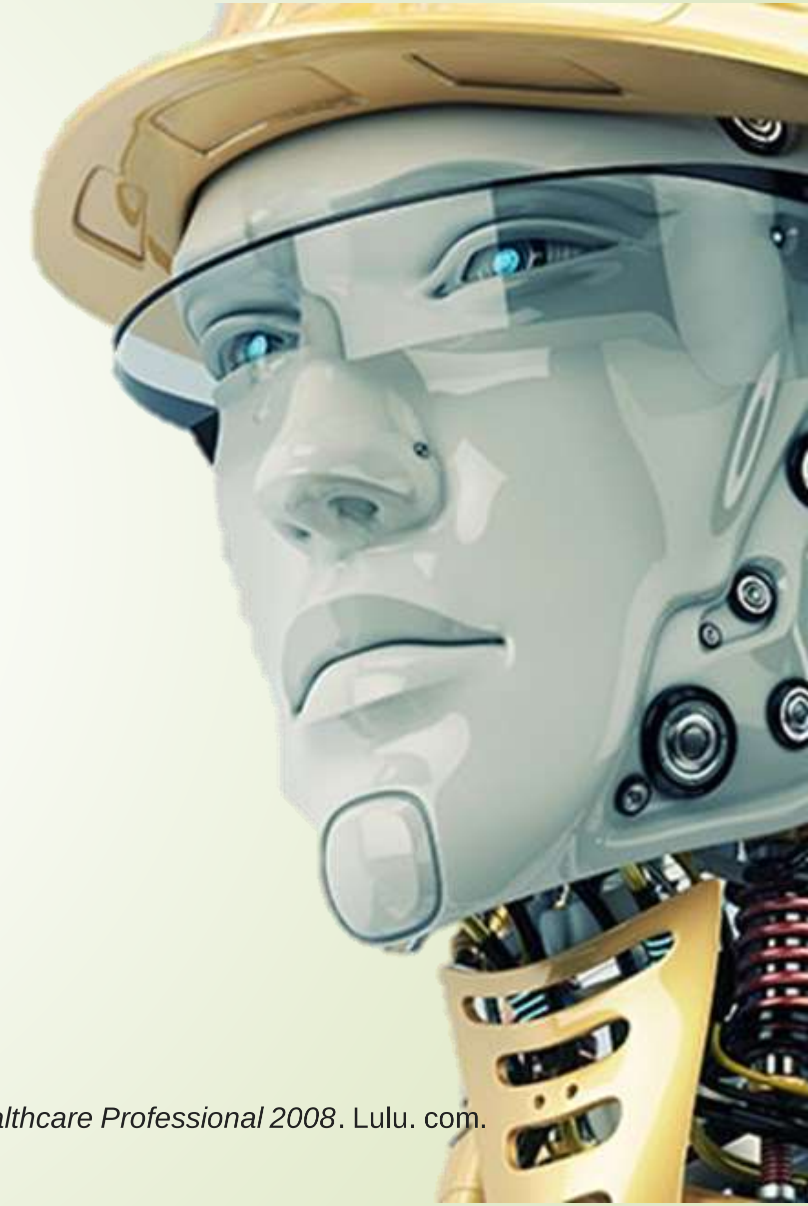
BACKGROUND

- As more medical information was published and more medical data was available (computerization) the need to automate, collect and analyze data arose.
- Also, as new technologies such as electronic health records appeared, ancillary technologies such as disease registries, voice recognition and picture archiving and communication systems arose to augment functionality.



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- Medical Informatics emphasizes sharing of a variety of information back and forth between people and healthcare entities.
  - **Examples of medical information** that needs to be shared:
    - lab results,
    - x-ray results,
    - vaccination status,
    - medication allergy status,
    - consultant's notes and hospital discharge summaries.
- Medical Informatics harnesses the **power of information technology to expedite the transfer and analysis of data**, leading to improved efficiencies and knowledge.

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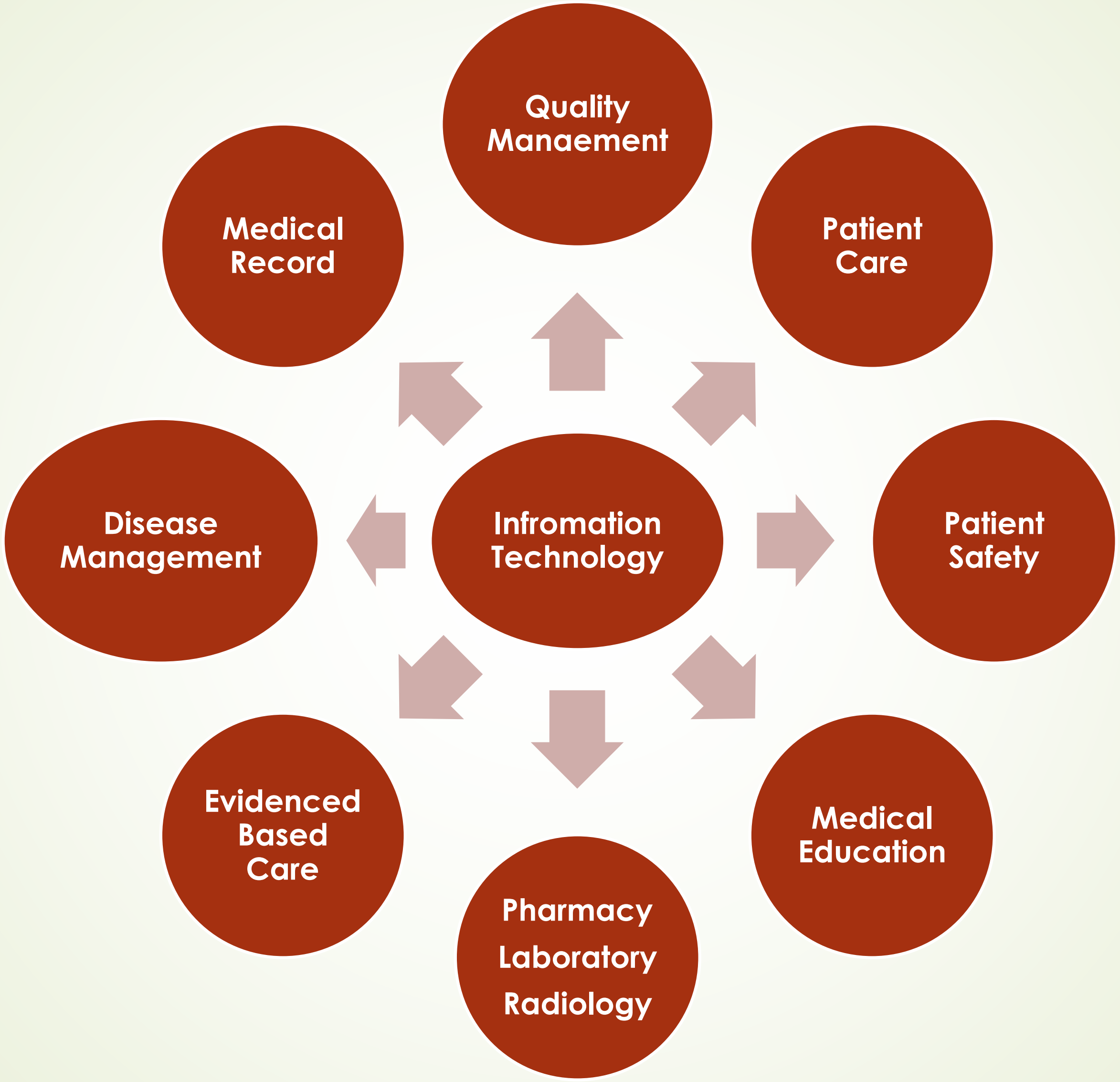
Aspects of Medical Informatics(Healthcare functions and information technology)



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- There are **multiple forces** driving the adoption of health information technology Like:
  - Decrease cost,
  - Improve patient safety and
  - Standardize and improve the quality of medical care

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Centerpiece of Medical Informatics

- The **electronic health record** (EHR) could be considered the centerpiece of Medical Informatics with its potential to improve patient safety, productivity and data retrieval.

- EHR use will eventually be shown to improve patient outcomes like morbidity and mortality as a result of decision support tools that decrease medication errors and standardize care with embedded clinical guidelines

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- The introduction of information technology into the practice of medicine has been **tumultuous/troubled for many reasons**:
  - new technologies expensive
  - physicians and nurses are frequently not adequately trained in technology
  - training rarely occurs in medical or nursing school or after graduation.

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Historical Highlights

Developments In Health Information Technology:

- **Computers**:
  - Useful for diagnosis and treatment; computers could archive and process information more rapidly than humans.
- **Origin of Medical Informatics**
- **MEDLINE:**
  - In the mid-1960's MEDLINE and MEDLARS were created to organize the world's medical literature
- **Artificial Intelligence.**
- **Internet**:
  - The Internet is the backbone for digital medical libraries, health information exchanges and web based medical applications, to include electronic health records
- **Electronic Health Record.**
- **Handheld technology**
  - Personal Digital Assistants (PDAs) loaded with medical software became standard equipment for any resident in training. They have been quickly supplanted by smartphones like the **iPhone**. PDAs and smartphones
- **Human Genome Project.**
  - collaborative research; To map all human genes was one of the greatest accomplishments in scientific history.
- **Nationwide Health Information Network**
  - The goal of the NHIN is to connect all electronic health records, regional networks and government agencies into one system, in one decade.

Key Players in Health Information Technology

- Information Technology is important to multiple players in the field of Medicine. The **common goals** are to:
  - Reduce medical errors
  - Provide better return on investment (ROI)
  - Improve communication and continuity among the key players
  - Improve the quality of care
  - Reduce duplication of tests or prescriptions ordered
  - Improve patient outcomes, like morbidity and mortality
  - Standardize care among clinicians, organizations and regions
  - Improve clinician productivity
  - Speed up access to care and administrative transactions
  - Protect privacy and ensure security

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Key Players In Health Information Technology

- the key players are: how they utilize health information technology
- **A. Patients:**
  - Online searches for health information
  - Web portals for storing personal medical information, making appointments, checking lab results, e-visits, etc
  - Research choice of physician, hospital or insurance plan
  - Online patient surveys
  - Online chat, blogs, podcasts, vodcasts and support groups and Web 2.0 social networking
  - Personal health records
  - Limited access to electronic health records
  - Telemedicine and home telemonitoring

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B. Clinicians:

- Online searches with MEDLINE, Google and other search engines
- Online resources and digital libraries
- Patient web portals, secure e-mail and e-visits
- Physician web portals
- Clinical decision support, e.g. reminders and alerts
- Electronic health records
- Personal Digital Assistants and smartphones loaded with medical software
- Telemedicine and telehomecare
- Voice recognition software
- Online continuing medical education
- Electronic (e)-prescribing
- Disease management and registries
- Picture archiving and communication systems
- Pay for performance
- Health Information Organizations
- E-research

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- **C. Nursing and support staff**
- Patient enrollment
- Electronic appointments
- Electronic billing process
- Electronic Health Records
- Web based credentialing
- Telehomecare monitoring
- Practice management software
- Secure patient-office e-mail communication
- Electronic medication administration record (e-Mar)
- Online educational resources and CME
- Disease registries

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- **D.Public Health**
  - Incident reports
  - Syndromic surveillance
  - Establish link to all public health departments (Public Health Information Network)
  - Geographic information systems to link disease outbreaks with geography
  - Telemedicine
  - Remote reporting using mobile technology

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- **E.Government**
  - Nationwide Health Information Network
  - Financial support
  - Information technology pilot projects and grants
    - o Disease management
    - o Pay for performance
    - o Electronic health records and personal health records

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Barriers To Health Information Technology (HIT) Adoption

- Inadequate time
- Cost
- Lack of interoperability
- Change in workflow.
- Privacy
- Legal
- Behavioral change.
- Inadequate workforce

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Summary?

Reference

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Thank You